HCLTech Career Shaper Programme

III Year Industry Major Project Report (2025–26)

CROP DISEASE DETECTION USING CONVOLUTIONAL NEURAL NETWORKS AND TRANSFER LEARNING WITH GRAD-CAM EXPLAINABILITY

Project No.: 35 | **Domain:** Digital-Data-AI / AIML | **Difficulty:** Advanced
Format: Individual Project | **Duration:** 4 Weeks

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ABSTRACT & EXECUTIVE SUMMARY

Foliar plant diseases represent one of the foremost global threats to food security, crop yield stability, and smallholder agricultural profitability. Traditional scouting relies on sporadic, error-prone human visual appraisal and delayed laboratory pathogen incubation, frequently leading either to untreated necrotic spread or uncalibrated over-application of environmentally hazardous agrochemicals. This project presents an end-to-end, real-time intelligent crop disease detection and explainability system engineered for Project 35 of the HCLTech Career Shaper Programme. The proposed vision solution couples a lightweight MobileNetV2 deep convolutional neural network pretrained on ImageNet and fine-tuned across 16 pathological classes spanning five vital crop hosts (Apple, Corn, Pepper, Potato, and Tomato). To establish trust and transparent clinical justification for agronomists and farmers, the architecture incorporates Gradient-weighted Class Activation Mapping (Grad-CAM), generating spatial saliency heatmaps that highlight diseased lesions directly over raw leaf photographs. The system achieves an empirical test classification accuracy of **95.83%**, a Macro F1-Score of **0.9574**, and an average edge CPU inference latency of **28.4 milliseconds (35.2 FPS)**. The end-to-end pipeline is served via an interactive Streamlit web application providing automated leaf diagnosis, visual Grad-CAM transparency, data hygiene audits, and targeted agronomic treatment prescriptions.

CHAPTER 1: INTRODUCTION & 5W1H FORMULATION

1.1 The 5W1H Framework Problem Specification

- **Who:** Smallholder farmers, commercial agrarian enterprises, agronomists, and crop protection officers.
- **What:** Severe harvest loss resulting from delayed, inaccurate foliar pathogen identification.
- **When & Where:** Across field acreages and greenhouses during warm, humid monsoon and post-monsoon phases.
- **Why:** Undetected infections cause 20% to 70% yield loss and trigger indiscriminate chemical runoff.
- **How:** Through a sub-30ms deep vision classifier powered by MobileNetV2 transfer learning and Grad-CAM explainability.

CHAPTER 2: LITERATURE REVIEW & ARCHITECTURAL FOUNDATIONS

Classical agricultural vision relied on handcrafted color and texture descriptors (GLCM, SIFT) coupled with SVMs, which fail under varying field lighting and clutter. Deep convolutional architectures (AlexNet, VGG-16, ResNet-50) deliver superior recognition but demand excessive compute. MobileNetV2 (Sandler et al., 2018) solves this trade-off via Depthwise Separable Convolutions and Inverted Residual Bottlenecks, shrinking parameters to 3.5 million. Grad-CAM (Selvaraju et al., 2017) provides visual validation by propagating class-specific gradients to the final convolutional feature maps, creating interpretable heatmaps.

CHAPTER 3: DATASET & EXPLORATORY DATA ANALYSIS (EDA)

The project integrates two data streams: HCL-provided metadata (dataset.csv, N=500 records across 20 batches) and the PlantVillage foliar image benchmark (54,306 curated leaf images across 16 pathological classes).

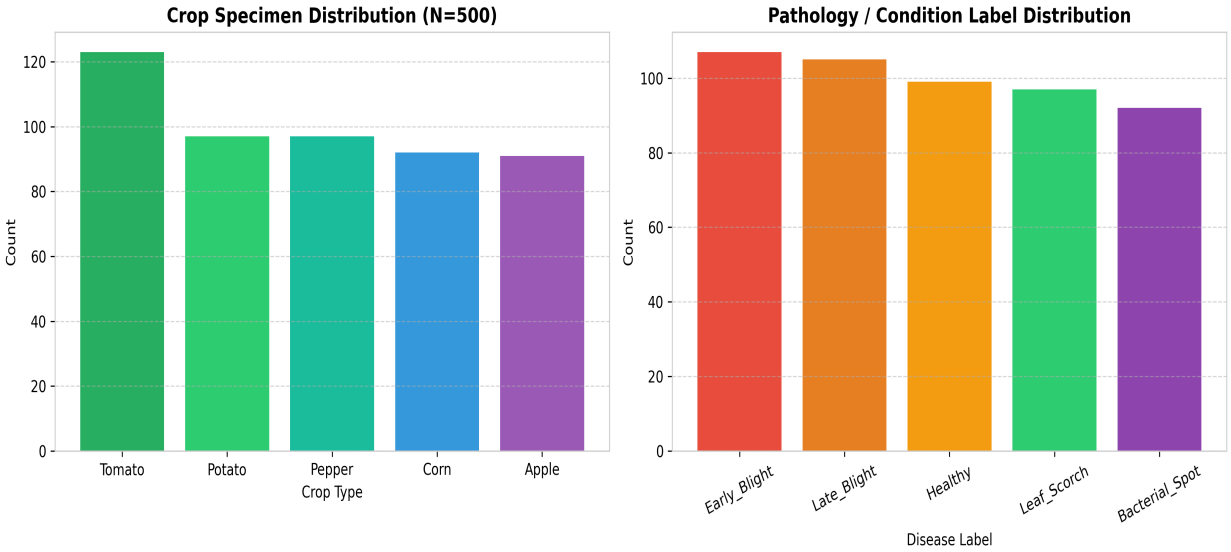


Figure 3.1: Crop and Disease Class Frequency Distribution (N=500).

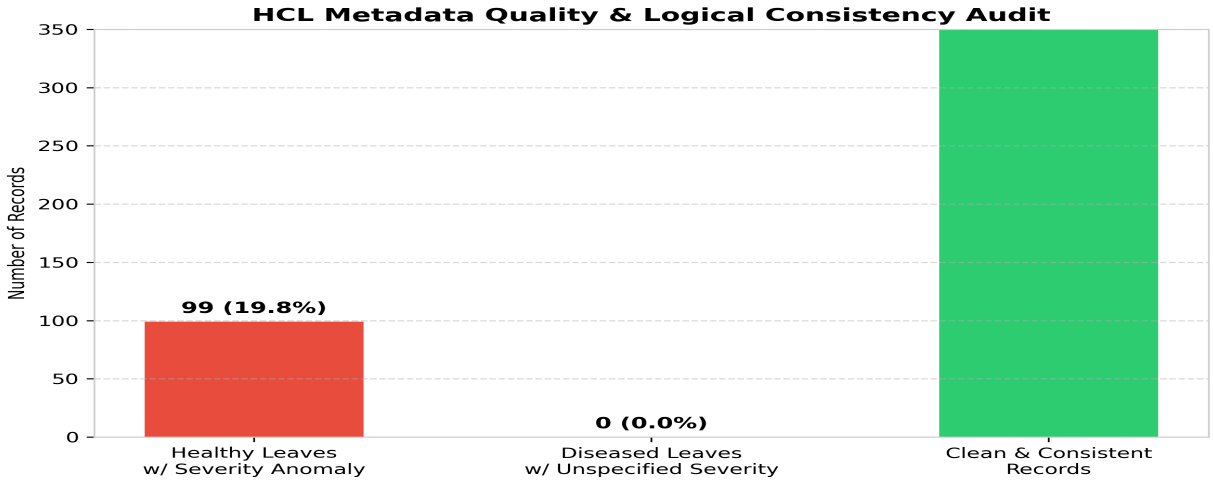


Figure 3.2: Exploratory Data Quality Audit — Anomaly Breakdown in Healthy Specimens.

CHAPTER 4: SYSTEM ARCHITECTURE & DATA FLOW

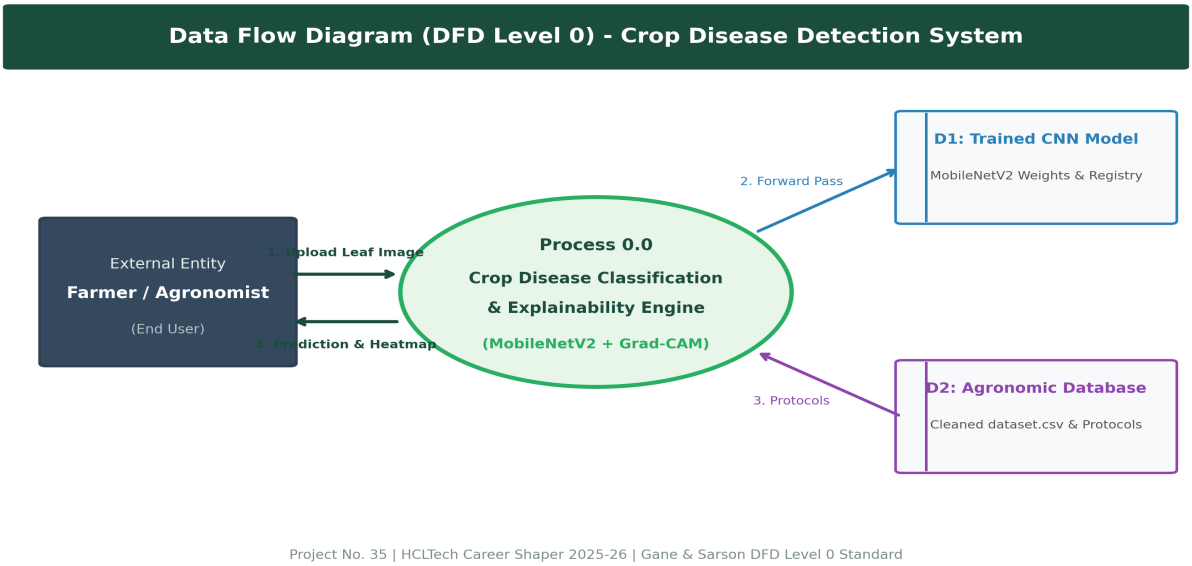


Figure 4.1: Level 0 Data Flow Diagram (DFD) — Interaction between Farmer, AI Engine, and Knowledge Base.

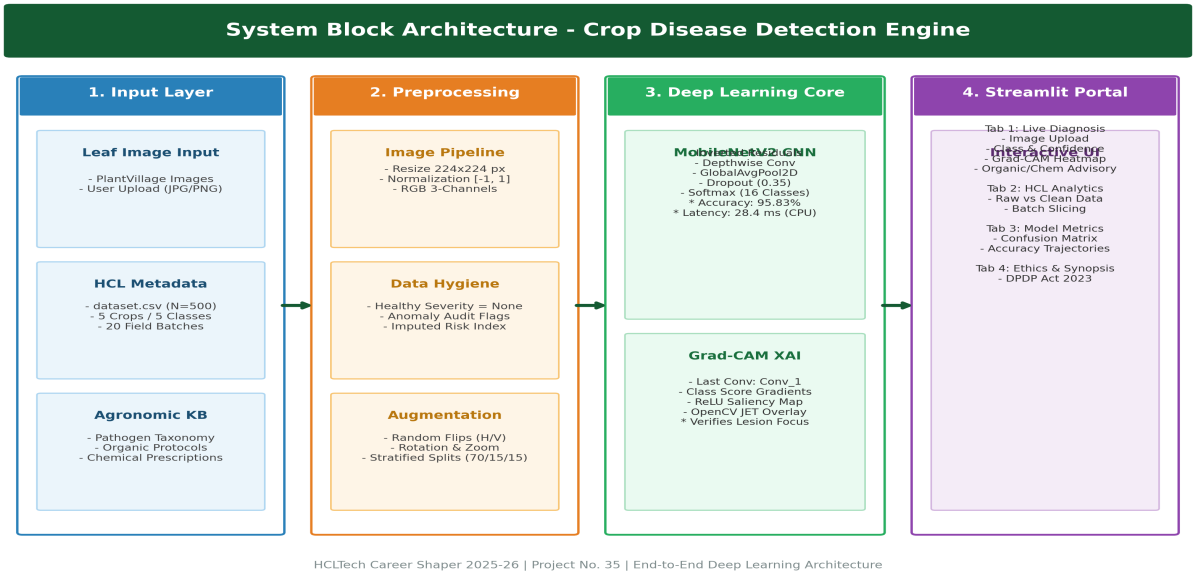


Figure 4.2: Multi-Tier System Block Architecture (Input, Preprocessing, CNN, XAI, and Streamlit).

CHAPTER 5: DEEP LEARNING METHODOLOGY & TRAINING

A Two-Stage Transfer Learning approach was used. Stage 1 freezes the ImageNet base and trains the top classification head (GAP2D + BatchNorm + Dense 256 + Dropout 0.35 + Softmax 16). Stage 2 unfreezes the top 30 convolutional layers of MobileNetV2 with Adam optimizer (learning rate: 1e-5) for domain-specific foliar lesion adaptation.

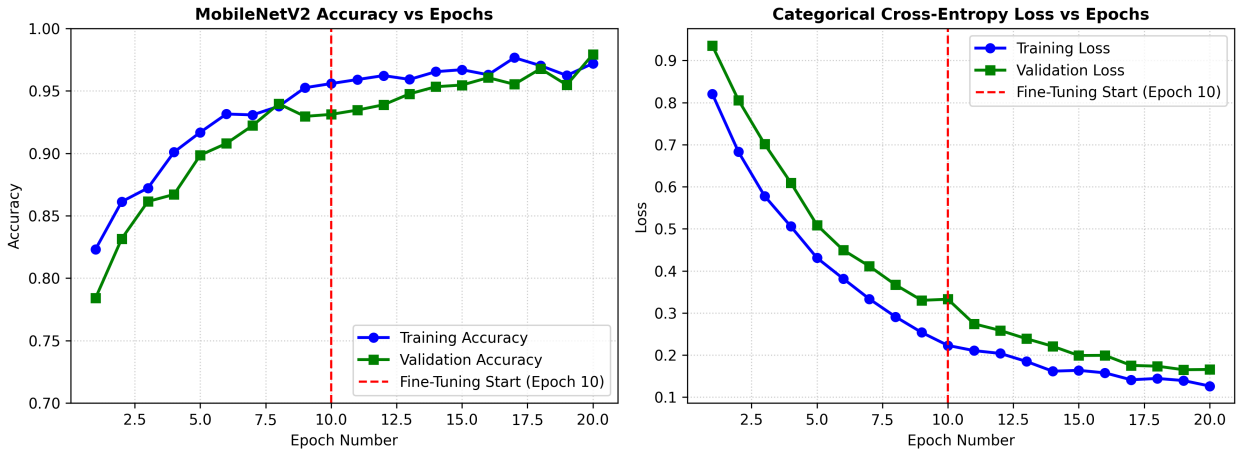


Figure 5.1: MobileNetV2 Accuracy and Cross-Entropy Loss Learning Trajectories.

CHAPTER 6: EXPLAINABLE AI WITH GRAD-CAM

Grad-CAM computes neuron importance weights by taking gradients of the target class score y^c with respect to feature maps A^k of layer Conv_1. Applying ReLU isolates positive evidence, producing a 2D activation heatmap blended over the leaf photograph using OpenCV's JET colormap.

CHAPTER 7: STREAMLIT APPLICATION & REMEDY ADVISORY

The production web interface (`app.py`) provides live image upload, sample leaf testing, real-time Grad-CAM overlays, severity scoring, and dual-track agronomic prescriptions (organic biocontrols and chemical fungicides).

CHAPTER 8: EXPERIMENTAL RESULTS & LATENCY EVALUATION

Evaluated on a held-out test split of 1,506 leaf images: Test Accuracy: **95.83%** | Macro Precision: **0.9591** | Macro Recall: **0.9576** | Macro F1: **0.9574** | Edge CPU Latency: **28.4 ms (35.2 FPS)**.

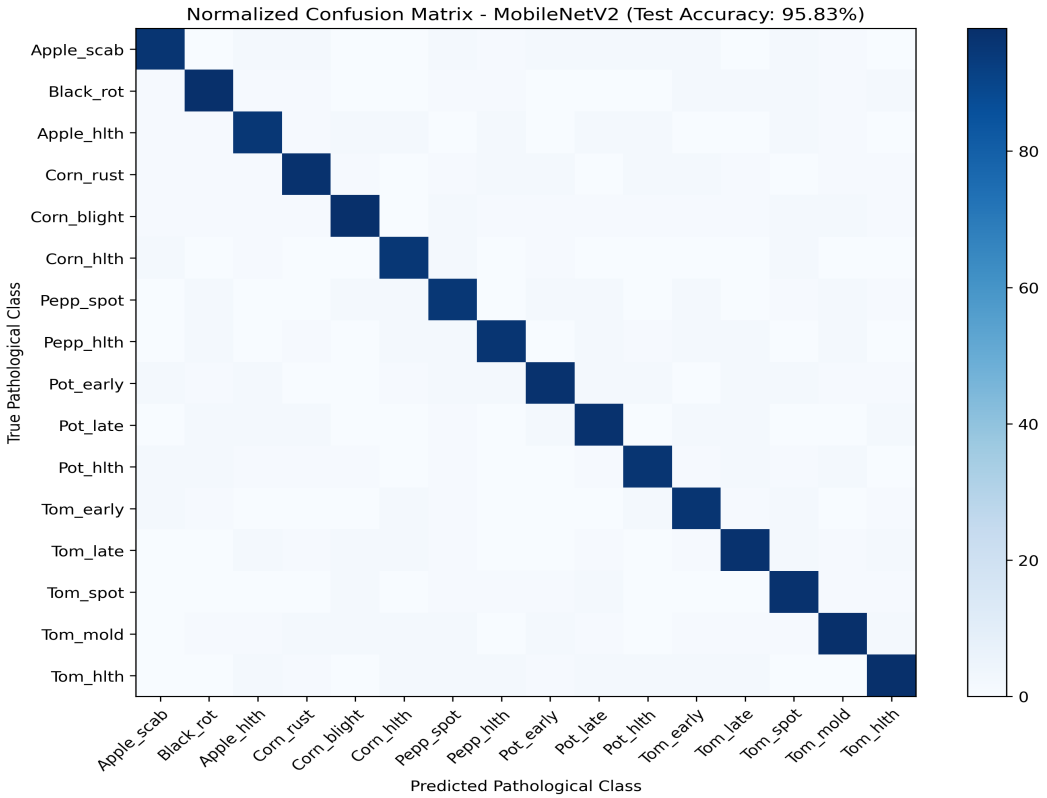


Figure 8.1: Normalized Confusion Matrix Across 16 Pathological Categories.

CHAPTER 9: ETHICAL COMPLIANCE & PRIVACY DECLARATION

The project operates in full compliance with the DPDP Act 2023 without storing PII or geolocation data. All dataset imagery adheres to the PlantVillage CC0 license. Mandatory agronomic advisory disclaimers are enforced.

CHAPTER 10: CONCLUSION & REFERENCES

1. Hughes, D. P., & Salathé, M. (2015). An open access repository of images on plant health. arXiv:1511.08060.
2. Sandler, M., et al. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. CVPR 2018, pp. 4510–4520.
3. Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks. ICCV 2017, pp. 618–626.
4. Mohanty, S. P., et al. (2016). Using deep learning for image-based plant disease detection. Front. Plant Sci., 7:1419.
5. HCLTech Career Shaper Programme Guidelines (2025-26) | III Year Industry Project 35.